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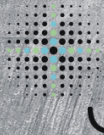
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Comprehensive Monte Carlo study of patient doses from cone-beam CT imaging in radiotherapy

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Abstract

Accurate knowledge of ionizing radiation dose from cone-beam CT (CBCT) imaging in radiotherapy is important to allow concomitant risks to be estimated and for justification of imaging exposures. This study uses a Monte Carlo CBCT model to calculate imaging dose for a wide range of imaging protocols for male and female patients. The Elekta XVI CBCT system was modeled using GATE and simulated doses were validated against measurements in a water tank and thorax phantom. Imaging dose was simulated in the male and female ICRP voxel phantoms for a variety of anatomical sites and imager settings (different collimators, filters, full and partial rotation). The resulting dose distributions were used to calculate effective doses for each scan protocol. The Monte Carlo simulated doses agree with validation measurements within 5% and 10% for water tank and thorax phantom respectively. Effective dose for head CBCT scans was generally lower for scans centred on the pituitary than the larynx (0.03 mSv versus 0.06 mSv for male ICRP phantom). Pelvis CBCT scan effective dose was higher for the female than male phantom (5.11 mSv versus 2.80 mSv for M15 collimator scan), principally due to the higher dose received by gonads for the female scan. Medium field of view thorax scan effective doses ranged from 1.38–3.19 mSv depending on scan length and phantom sex. Effective dose for half rotation thorax scans with offset isocentre varied by almost a factor of three depending on laterality of the isocentre, patient sex and imaged field length. The CBCT imaging doses simulated here reveal large variations in dose depending on imaging isocentre



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location, patient sex and partial rotation angles. This information may be used to estimate risks from CBCT and to optimize CBCT imaging protocols.

Keywords: cone-beam CT, image-guided radiotherapy, Monte Carlo simulation, imaging dose

(Some figures may appear in colour only in the online journal)

1. Introduction

Cone beam CT (CBCT) imaging is commonly used in radiotherapy for verification of patient and target position [1, 2]. Use of CBCT imaging can improve the quality of radiotherapy by reducing geometric errors, leading to the use of smaller margins, and providing potential for adaptive radiotherapy. However use of CBCT is not without risk to the patient due to the additional dose of ionizing radiation involved. Accurate knowledge of patient dose from CBCT is therefore important so that concomitant risks can be estimated allowing justification of additional imaging. This is particularly important because CBCT imaging is typically repeated, sometimes at every fraction of treatment, leading to potentially large total doses.

Imaging doses from CBCT in radiotherapy have been reported in many studies since its introduction in the early 2000s [3]. Reported doses are either based on direct measurements, or on modeling often using Monte Carlo (MC) methods. Dose measurements are most commonly made in phantoms, which may represent patient anatomy to a greater or lesser extent [4–9]. Measurements are time consuming to perform due to the large number of measurement points required to represent the full distribution of dose in the patient. Effective dose estimates, which allow the linking of dose to risk, require the mean absorbed dose to each organ (subsequently referred to as ‘organ dose’) to be measured as defined in ICRP103 [10]. Some studies [6, 7, 11–13] have calculated effective doses for CBCT based on dose measurements at the various organ locations, however this involves uncertainty due to the non-uniformity of dose within each organ [14].

Monte Carlo methods have been used to overcome the limitations of phantom measurements [14–19]. MC offers the flexibility to simulate full 3D dose distributions within a range of objects including voxel phantoms based on actual patient anatomy. However, it can be time consuming to achieve acceptable statistical errors with MC simulations, and the models must be validated to ensure they accurately reproduce doses within complex patient anatomy.

Most studies of CBCT dose have reported doses for a relatively small number of standard imaging protocols—e.g. head, thorax and pelvis. In clinical practice CBCT imaging is applied at wide range of anatomical sites and there are many imaging protocol variations, for example different fields of view (FOVs), filters, full and half rotation scans. The effect of these variations on CBCT imaging dose has not been comprehensively studied. Also CBCT doses in the literature are usually reported only for male patients. Some quote doses based on female patient image sets or phantoms [11, 13, 17, 20–23], but few compare male and female doses from the same imaging protocol [24].

In this study we have developed an MC model of the Elekta XVI CBCT imaging system (Elekta AB, Stockholm, Sweden). We use this model to calculate organ and effective doses in the ICRP male and female voxel phantoms [25] for a wide range of imaging sites (including male and female, brain, larynx, lung (left, right and central), prostate, cervix, breast (left and right)) and scanning protocols (including small, medium and large FOV, with and without bow-tie filter, full and half rotation scans with various angular ranges). The ICRP

computational phantoms were previously used in dosimetry studies of Varian Truebeam linac-integrated CBCT [26] and of dental CBCT [27].

2. Methods and materials

2.1. Monte Carlo model of Elekta XVI CBCT

The Elekta XVI CBCT system was modeled using the Geant4 application for tomographic emission (GATE) [28]. GATE has previously been used to model a wide range of medical physics applications including proton therapy, megavoltage x-ray therapy and kilovoltage x-ray CBCT [18, 29]. The results shown in this report were calculated using GATE v7.1 (March 2015), which is based on Geant4 v10.1. The Geant4 ‘physics list’ mechanism [30] is used to specify processes to be included in the simulation. All simulations were done using the EMLivermore physics list, which is optimized for accuracy of electron and photon physics in the low (kilovoltage) energy range. This physics list has been used and recommended in other studies of kV x-ray imaging/therapy [31, 32]. The GATE XVI model includes the following components:

2.1.1. X-ray source. A ‘virtual’ x-ray source has been defined using the general particle source object in GATE. This allows an x-ray source of arbitrary shape with user-defined energy spectrum to be created. Energy spectra corresponding to peak tube voltages of 100kV and 120kV with filtration equivalent to 5.25 mm Al were generated according to the methods described in Boone 1997 [33]. The source has been defined as a rectangle of size 0.8 mm $\times$ 0.4 mm, to match the size of the focal spot given in the XVI system documentation. The rectangular source is embedded a small distance (2.5 μ m) below the surface of a tungsten disc, representing the generation of x-rays close to the surface of the anode in the x-ray tube. The depth of the source within the anode reproduces the differential self-absorption of x-rays within the anode as a function of emission angle (anode-heel effect), with the depth used adjusted empirically to best match dose profiles across the beam measured in water (see section 2.2.1). Simulations of 120 keV electrons incident on a tungsten target showed that 2.5 μ m is close to the mean depth of interaction. The tungsten target with embedded photon source is rotated at an angle of 17.5° from the beam axis to simulate the anode angle of the x-ray tube (14°) plus the tilt of the x-ray tube mounting relative to the central beam axis. The centre of the photon source is located at 100 cm from the centre of rotation of the CBCT system (nominal linac isocentre).

2.1.2. Collimators. The XVI system has a series of interchangeable collimator cassettes each containing a lead sheet with cut-outs of different size and shape to define the different x-ray field sizes available. Collimators are identified by a letter (S, M or L) indicating the reconstructed FOV diameter, and a number (10, 15, 20) indicating the FOV length. Nominal FOV sizes for the different collimators are given in Spezi 2009 [34]. Each collimator is modeled as a 3.2 mm thick sheet of lead with appropriate cutouts. Small adjustments (<1 mm) were made to the nominal cut-out dimensions and offsets to best match measured profiles. Collimators are positioned at a distance of 20 cm from the x-ray source along the beam axis towards the centre of rotation. The collimator cassettes also have thin plastic covers on either side which are included in the model as 1 mm thicknesses of PET-G.

2.1.3. Bow tie filter. The XVI CBCT system contains a removable aluminium (Al2017 alloy) bow tie filter (referred to as F1). The F1 filter was modeled according to the specifications in the XVI documentation, with the curved shape of the filter approximated in GATE as a series

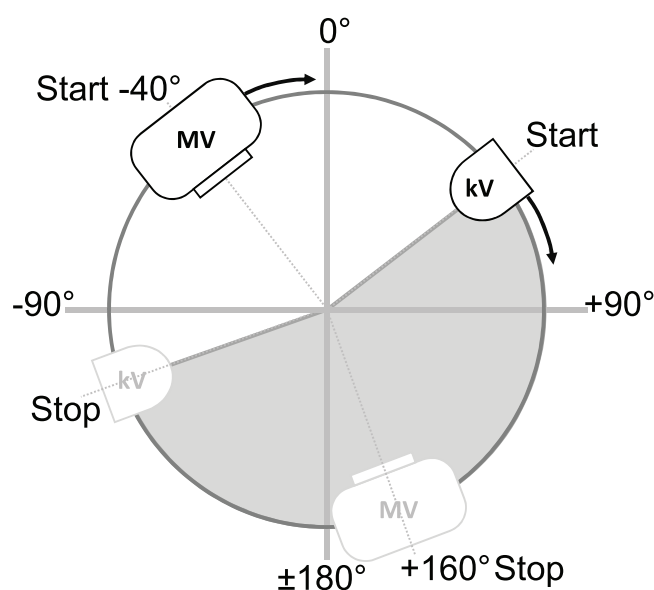


Figure 1. Rotation of the XVI gantry (view in direction of gantry). Marked gantry angles indicate the position of the MV linac head. Start and stop positions for H&N scan half rotation are shown with the shaded segment indicating angles traversed by the kV beam.

of 100 trapezoids. The centre of the bow tie filter is located at a distance of 25 cm from the x-ray source. The bow tie filter may be removed and replaced with an empty filter cassette, referred to as F0.

2.1.4. Gantry rotation. Rotation of the CBCT gantry has been modeled in GATE through rotation of the patient volume (containing the representation of the patient or phantom being exposed). Rotations are applied about an axis running along the length of the treatment couch and passing through the linac isocentre. This allows the patient volume to be exposed from any gantry angle. In practice the rotation speed of the XVI system is not constant throughout the CBCT acquisition. Initial projection images are acquired with the gantry static in the start position, which then accelerates to a constant speed for most of the rotation, but then decelerates to a stop over the final few degrees of the rotation. Full rotation acquisitions were modeled in GATE as projections from 200 different gantry angles sampled at equal time increments from a measured gantry angle trajectory. A similar rotation trajectory was measured for half rotation scans and used to model these acquisitions, also as 200 discrete projection angles. Figure 1 illustrates rotation of the gantry for the head and neck (H&N) half scan protocol starting at -40° and finishing at $+160^\circ$. Specified gantry angles always refer to the position of the megavoltage (MV) linac head with 0° being directly above the patient couch. The gantry can rotate in either direction to $\pm 180^\circ$, but cannot rotate past 180° . The kV tube is mounted on the gantry with an offset of $+90^\circ$ from the linac head as shown in figure 1.

2.1.5. Dose output from GATE. Dose output in GATE is facilitated using ‘Dose Actors’ which can be defined as a 2D or 3D arrays covering a region in space or to coincide with the voxel locations of an image based phantom. The absorbed dose to the material in each voxel is returned by the dose actor. Simulation times in GATE were reduced by using the split

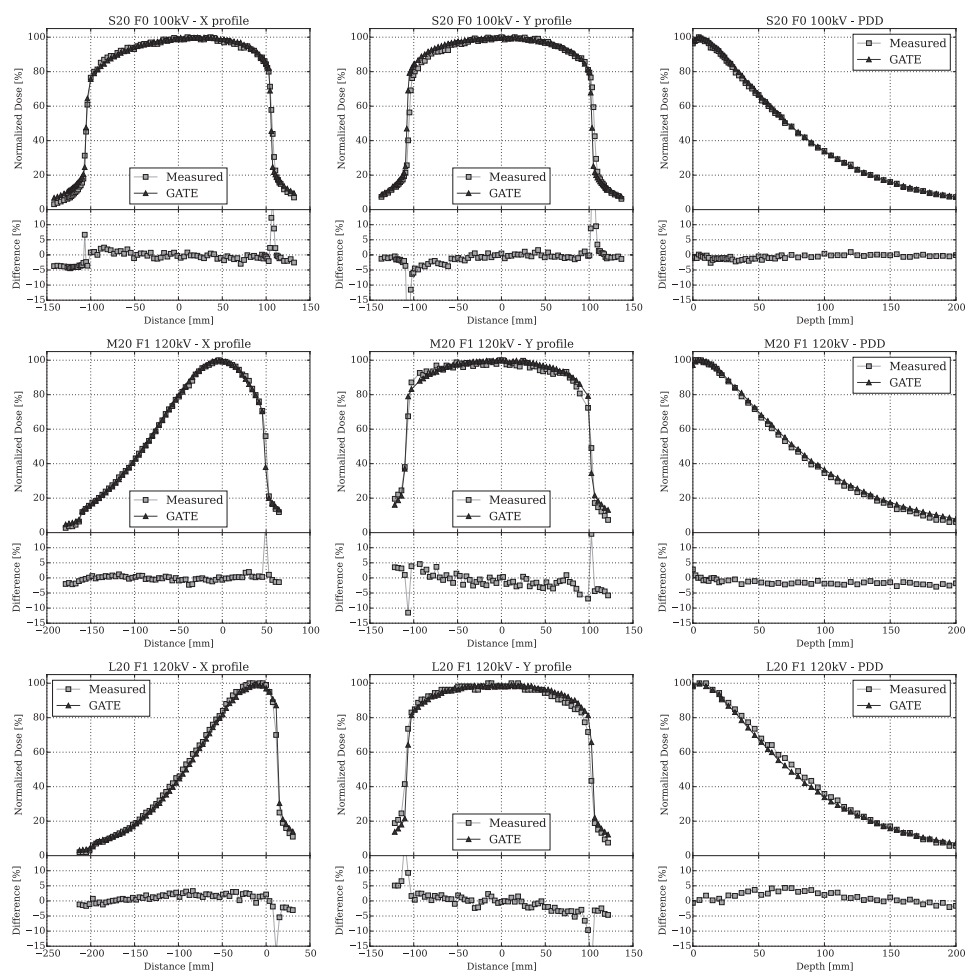


Figure 2. Comparison of simulated and measured dose profiles in a water tank.

exponential track length estimator dose actor (seTLE, introduced in GATE7.1), which makes use of trajectory splitting and exponential track-length estimator methods [35] to improve the efficiency of simulations.

2.2. Model validation

The GATE XVI model has been validated using dose measurements in a water tank and in an anthropomorphic thorax phantom.

2.2.1. Water tank profiles. Three-dimensional dose distributions have been measured in a water-filled plotting tank (PTW MP3 tank) using a small volume ionization chamber (PTW Semiflex 31002). The gantry was rotated to -90° to position the kV source directly above the water tank with the beam aiming downwards into the water, and with kV source to surface distance of 75 cm. X, Y and depth profiles were acquired for each collimator with the F0 and F1 filters and at 100 and 120 kV. Profiles were scanned with a spatial resolution of

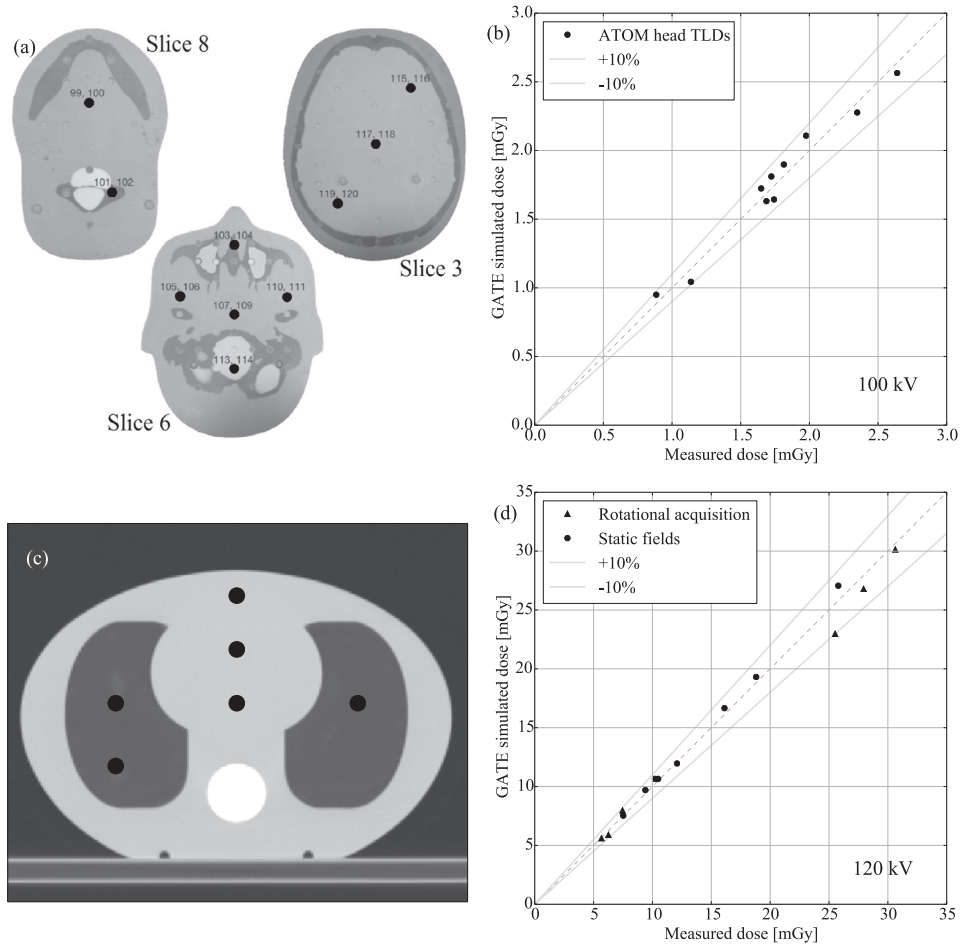


Figure 3. (a) Dose measurement locations in the ATOM head phantom (black circles). (b) Comparison of measured and simulated doses in the ATOM head phantom. (c) Dose measurement locations in the CIRS thorax phantom (black circles). (d) Comparison of measured and simulated doses in the CIRS thorax phantom.

2 mm between measurement points in the beam penumbra and build-up regions and 5 mm between points elsewhere. A measurement time of 2 s per point was used. The x-ray tube was set to deliver 1.6 mAs per frame and 5 frames per second. The plotting tank was modeled in GATE as a cube of water and dose profiles corresponding to those measured were simulated with a spatial resolution of 2 mm. The GATE profiles were normalized and compared to the measurements.

2.2.2. Model calibration. Dose measurements were made in head and thorax anthropomorphic phantoms in order to calibrate the MC simulated doses, using a procedure similar to that of Ding *et al* [15]. These measurements were used to derive a Monte Carlo calibration factor f_{MC} :

$$D = f_{MC} D_{MC}$$

Table 1. Organ and effective doses simulated in the ICRP phantoms for head and neck (H&N) CBCT scans. ^aHalf rotation -40° to $+160^\circ$.

Preset	H&N S10	H&N S10	H&N S10	H&N S10	Fast H&N S20	Fast H&N S20	Fast H&N S20	Fast H&N S20
Isocentre	Pituitary	Pituitary	Larynx	Larynx	Pituitary	Pituitary	Larynx	Larynx
Sex M/F	F	M	F	M	F	M	F	M
kV	100	100	100	100	100	100	100	100
Collimator	S10	S10	S10	S10	S20	S20	S20	S20
Filter	F0	F0	F0	F0	F0	F0	F0	F0
Total mAs	36.6	36.6	36.6	36.6	18.3	18.3	18.3	18.3
Rotation	half ^a	half ^a	half ^a	half ^a	half ^a	half ^a	half ^a	half ^a
<i>Organ doses</i>								
<i>(mGy)</i>								
Bone marrow (red)	0.13	0.08	0.11	0.07	0.11	0.06	0.12	0.07
Colon	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Lung	0.01	0.00	0.08	0.06	0.01	0.01	0.16	0.12
Stomach	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.01
Breast	0.00	0.00	0.02	0.01	0.01	0.00	0.07	0.01
Gonads	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bladder	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oesophagus	0.02	0.01	0.21	0.19	0.07	0.04	0.18	0.14
Liver	0.00	0.00	0.01	0.01	0.00	0.00	0.01	0.01
Thyroid	0.04	0.01	0.55	0.56	0.20	0.05	0.32	0.31
Bone surface	0.44	0.42	0.33	0.26	0.33	0.29	0.38	0.30
Brain	0.63	0.62	0.03	0.01	0.39	0.37	0.13	0.06
Salivary glands	0.66	0.56	0.47	0.26	0.48	0.46	0.48	0.45
Skin	0.05	0.05	0.06	0.05	0.05	0.04	0.07	0.06
Remainder	0.07	0.06	0.10	0.05	0.06	0.05	0.09	0.07
Eye lens	0.59	0.56	0.03	0.01	0.30	0.28	0.29	0.04
Effective dose (mSv)	0.05	0.03	0.08	0.06	0.05	0.03	0.09	0.06

which converts simulated dose per photon, D_{MC} , into dose delivered by the CBCT system per mAs, D . The dose per mAs can then be scaled by the total mAs for the CBCT protocol to give the dose for that scan. Separate MC calibration factors are required for each simulated source energy (kVp).

Calibration measurements for the 100kV beam were made using thermoluminescent dosimeters (TLDs) (Harshaw TLD-100H $3 \times 3 \times 1$ mm chips) at various locations in a head phantom (CIRS ATOM 702), using a clinical head and neck scan exposure (S20 collimator). TLD measurement locations in the head phantom are shown in figure 3(a), with two TLD chips being placed at each location. Measurements for the 120kV beam were made in a thorax phantom (CIRS 002LFC) using a small volume ionization chamber (PTW semiflex 31002). Thorax phantom measurements included both static beam angle and rotational exposures using the M20 chest preset. Measurement locations in the thorax phantom are shown in figure 3(c). Similar to previous studies [15, 36], anthropomorphic phantoms were used for the calibration measurements because they allow clinical scan protocols to be used, including rotational acquisition and realistic patient geometry. Both TLDs and the small volume

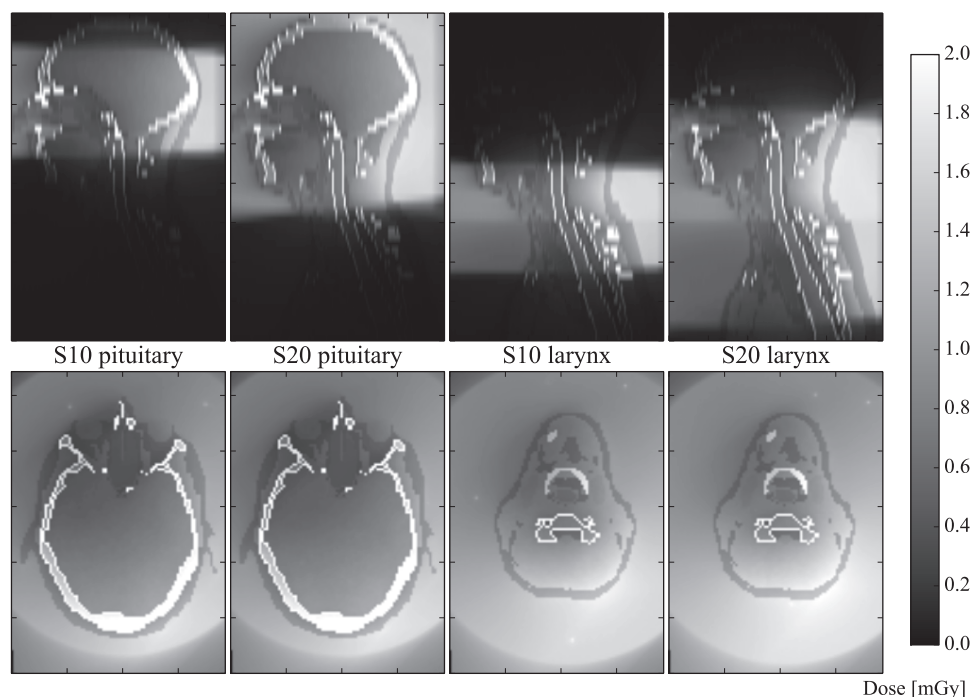


Figure 4. Simulated dose distributions in the ICRP male phantom for CBCT scans of head and neck region.

ionization chamber were first calibrated in the XVI beam by inter-comparison in air with a Radcal 10×6–6 cc chamber (itself calibrated traceable to national standards).

Dose measurements at each location in the phantom were compared to GATE simulations of the same exposures, using voxelized representations of the same phantoms (CT scans). The MC calibration factor at each kV was determined by minimizing the mean difference between measurements and simulations. Variations of the dose differences at each location in the phantom were also used to validate the simulated dose distribution within an anthropomorphic phantom containing materials of different tissue types (lung, soft-tissue, bone) from rotational exposures.

2.3. Simulation of patient CBCT doses

Dose from a full range of CBCT imaging protocols was simulated using the ICRP male and female computational phantoms. The ICRP phantoms are based on medical image data of real people yet are consistent with reference anatomical parameters given in ICRP report 89 [37]. The voxel phantoms consist of an organ type and material composition specified at each voxel. The voxel phantom was imported into the GATE simulation with the appropriate material specified at each voxel. For each CBCT protocol simulated 10^7 primary x-rays were generated, and doses were recorded using the seTLE dose actor with primary and secondary multiplicities of 200 and 400 respectively. The full phantom body length was included in each simulation and the dose scoring array was defined to cover the whole phantom with a voxel

Table 2. Organ and effective doses simulated in the ICRP phantoms for pelvis CBCT scans. ^bHalf rotation: gantry angle -130° to $+70^{\circ}$.

Preset	Prostate M10	Prostate M15	Pelvis M15	Pelvis M15	Pelvis M20	Pelvis M20	Pelvis L20	Pelvis L20	Fast Prostate Seed S10
Isocentre	Prostate	Prostate	Uterus	Prostate	Uterus	Prostate	Uterus	Prostate	Prostate
Sex M/F	M	M	F	M	F	M	F	M	M
kV	120	120	120	120	120	120	120	120	120
Collimator	M10	M15	M15	M15	M20	M20	L20	L20	S10
Filter	F1	F1	F1	F1	F1	F1	F1	F1	F0
Total mAs	1689.6	1689.6	1056.0	1056.0	1056.0	1056.0	1689.6	1689.6	46.8
Rotation	full	full	full	full	full	full	full	full	half ^b
<i>Organ doses (mGy)</i>									
Bone marrow (red)	5.30	6.52	4.76	4.08	6.73	5.82	7.88	6.98	0.29
Colon	2.23	3.72	8.84	2.33	15.45	4.26	17.03	4.53	0.09
Lung	0.09	0.11	0.10	0.07	0.17	0.11	0.23	0.15	0.00
Stomach	0.15	0.20	0.24	0.13	0.53	0.22	0.68	0.28	0.00
Breast	0.11	0.13	0.09	0.08	0.15	0.13	0.20	0.19	0.00
Gonads	6.46	13.91	22.11	8.70	24.52	31.22	27.49	34.34	0.36
Bladder	28.02	35.12	27.98	21.95	29.98	25.48	34.63	29.83	1.39
Oesophagus	0.06	0.08	0.07	0.05	0.13	0.08	0.17	0.13	0.00
Liver	0.14	0.19	0.21	0.12	0.45	0.24	0.56	0.31	0.00
Thyroid	0.07	0.08	0.10	0.05	0.13	0.08	0.16	0.12	0.00
Bone surface	11.52	14.69	9.91	9.18	15.01	14.07	18.02	16.95	0.60
Brain	0.04	0.05	0.04	0.03	0.06	0.05	0.08	0.07	0.00
Salivary glands	0.08	0.09	0.06	0.06	0.10	0.08	0.12	0.13	0.00
Skin	3.43	4.61	3.26	2.88	5.12	4.70	6.41	5.86	0.19
Remainder	3.30	3.94	3.22	2.46	4.27	3.24	5.06	4.00	0.15
Effective dose (mSv)	3.14	4.48	5.11	2.80	6.67	5.37	7.60	6.13	0.16

resolution matching the ICRP phantom definition ($2.1 \times 2.1 \times 8.0$ mm for the male phantom, $1.8 \times 1.8 \times 4.8$ mm for the female phantom). Mean organ doses were calculated by averaging the dose in all voxels belonging to each organ. Organs with multiple parts (e.g. left and right breast) were considered to be part of a combined organ for the purposes of calculating the mean dose.

Doses were simulated from all default imaging protocols in XVI version 5.0. H&N protocols were simulated with both pituitary and larynx positioned at the isocentre, to represent the range of different locations at which the H&N protocol may be applied. Pelvis protocols were centred on prostate for the male phantom and on uterus for the female phantom. Additional protocols for lung and breast imaging that are used in our institution were also simulated. Half rotation lung protocols (Small FOV) are used in cases where the isocentre is laterally offset from the patient midline, preventing full rotation of the gantry around the patient. Different gantry start and stop angles are required depending on the laterality of the isocentre, and the isocentre location used was the centre of either the left or right lung. Large FOV lung protocols were also simulated, as well as half rotation protocols for left and right breast imaging. In total 41 CBCT image acquisitions were simulated.

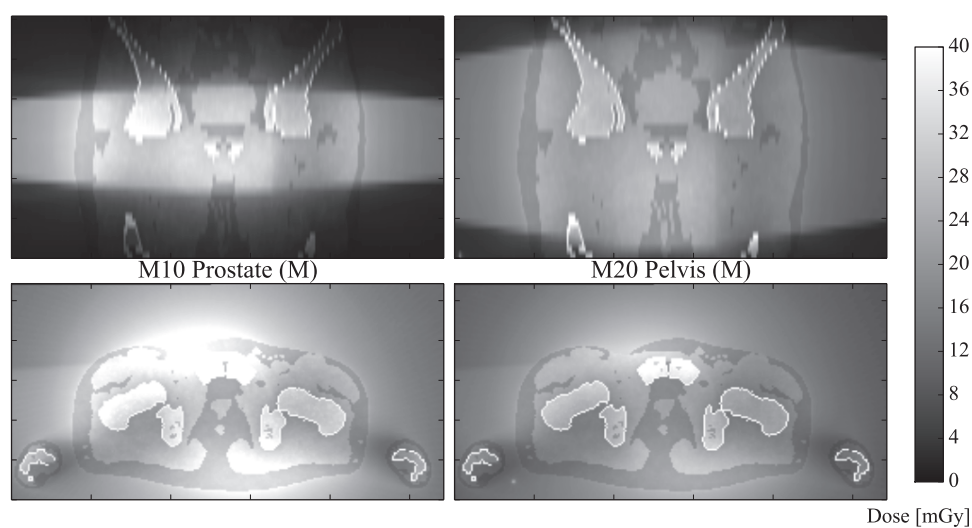


Figure 5. Simulated dose distributions in the ICRP male phantom for pelvis CBCT scans.

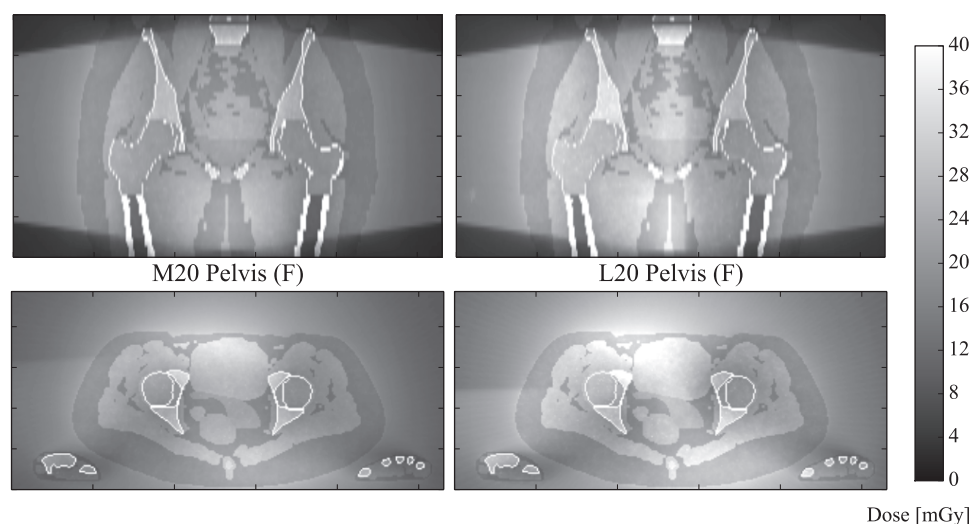


Figure 6. Simulated dose distributions in the ICRP female phantom for pelvis CBCT scans.

3. Results

3.1. Model validation and calibration

Dose distributions in the water tank showed good agreement between simulations and measurements, with differences less than 5% in all areas excluding beam penumbras. Figure 2 shows sample profiles for S20, M20 and L20 collimators. Profiles for other collimators both with and without the bow tie filter showed similar levels of agreement.

Table 3. Organ and effective doses simulated in the ICRP phantoms for MFOV and 4D chest CBCT scans. ^cHalf rotation: gantry angle -180° to $+20^{\circ}$.

Preset	Chest M10	Chest M10	Chest M15	Chest M15	Chest M20	Chest M20	Symmetry 4D	Symmetry 4D
Isocentre	Lung centre	Lung centre	Lung centre	Lung centre	Lung centre	Lung centre	Lung centre	Lung centre
Sex M/F	F	M	F	M	F	M	F	M
kV	120	120	120	120	120	120	120	120
Collimator	M10	M10	M15	M15	M20	M20	S20	S20
Filter	F1	F1	F1	F1	F1	F1	F0	F0
Total mAs	264.0	264.0	264.0	264.0	264.0	264.0	312.0	312.0
Rotation	full	full	full	full	full	full	half ^c	half ^c
<i>Organ doses (mGy)</i>								
Bone marrow (red)	0.76	0.63	1.04	0.88	1.69	1.45	3.53	3.20
Colon	0.04	0.12	0.06	0.18	0.12	0.48	0.21	1.30
Lung	4.85	3.81	5.82	4.56	6.76	5.36	14.04	13.22
Stomach	0.67	0.75	1.16	1.37	3.26	3.27	8.66	9.98
Breast	4.76	3.72	5.15	4.13	5.45	4.41	22.18	19.77
Gonads	0.01	0.01	0.02	0.02	0.03	0.03	0.03	0.03
Bladder	0.01	0.01	0.02	0.02	0.03	0.03	0.04	0.03
Oesophagus	2.92	2.32	3.95	3.10	5.92	4.61	10.75	8.87
Liver	0.90	0.88	1.72	1.53	4.37	3.61	10.46	9.13
Thyroid	0.83	0.69	1.39	1.10	8.80	7.68	21.58	19.19
Bone surface	2.42	2.27	3.23	3.02	4.93	4.61	8.70	8.19
Brain	0.04	0.03	0.06	0.04	0.12	0.09	0.18	0.12
Salivary glands	0.18	0.12	0.26	0.17	0.57	0.37	1.26	0.79
Skin	0.65	0.51	0.84	0.67	1.26	1.03	3.12	2.48
Remainder	1.05	0.87	1.46	1.24	2.37	2.12	5.18	4.79
Effective dose (mSv)	1.67	1.38	2.09	1.75	3.19	2.75	8.30	7.88

MC calibration factors, f_{MC} , derived from the head and thorax phantom measurements/simulations were 2.61×10^{11} photons per mAs at 100 kV, and 5.63×10^{11} photons per mAs at 120 kV. Dose distributions in the head phantom at 100 kV and in the thorax phantom at 120 kV showed good agreement with GATE simulations, shown in figures 3(b) and (d). The difference between measured and simulated dose was less than 10% at all points (range -8% to $+7\%$ for head phantom, -9% to $+7\%$ for thorax phantom) with standard deviation of 5.4% for head phantom and 4.6% for the thorax phantom.

3.2. Simulation of patient CBCT doses

Each simulation of CBCT scan dose in the ICRP phantom using 10^7 primary photons took approximately 2 h to simulate using an AMD Opteron system with 64 2.3 GHz cores. Repeat simulations showed statistical uncertainty in effective dose estimates below 1%.

Table 1 shows the organ and effective doses simulated in the ICRP phantom for head and neck CBCT scans. Effective doses are higher for scans with the S20 collimator than the S10 collimator due to the longer field length, although this is mitigated by the lower total mAs used for the 'Fast' S20 head and neck protocols. Effective doses are also higher for scans centred

Table 4. Organ and effective doses simulated in the ICRP phantoms for SFOV chest CBCT scans. ^cHalf rotation: gantry angle -180° to $+20^{\circ}$. ^dHalf rotation: gantry angle -80° to $+120^{\circ}$.

Preset	Left lung S10	Left lung S10	Left lung S20	Left lung S20	Right lung S10	Right lung S10	Right lung S20	Right lung S20
Isocentre	L lung	L lung	L lung	L lung	R lung	R lung	R lung	R lung
Sex M/F	F	M	F	M	F	M	F	M
kV	120	120	120	120	120	120	120	120
Collimator	S10	S10	S20	S20	S10	S10	S20	S20
Filter	F1	F1	F1	F1	F1	F1	F1	F1
Total mAs	117.1	117.1	117.1	117.1	117.1	117.1	117.1	117.1
Rotation	half ^d	half ^d	half ^d	half ^d	half ^c	half ^c	half ^c	half ^c
<i>Organ doses (mGy)</i>								
Bone marrow (red)	0.36	0.31	0.82	0.74	0.36	0.32	0.83	0.74
Colon	0.02	0.07	0.05	0.33	0.03	0.08	0.08	0.29
Lung	2.23	1.67	3.06	2.34	2.48	2.20	3.34	2.99
Stomach	0.46	0.52	2.49	2.39	0.36	0.44	1.74	1.85
Breast	1.92	1.69	2.18	1.93	4.10	3.45	4.53	3.97
Gonads	0.01	0.01	0.02	0.01	0.01	0.01	0.01	0.02
Bladder	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02
Oesophagus	1.36	1.03	2.57	1.92	1.26	1.06	2.66	2.15
Liver	0.31	0.23	1.25	0.71	0.63	0.63	2.84	2.48
Thyroid	0.39	0.31	2.84	2.54	0.44	0.38	5.15	4.47
Bone surface	1.27	1.20	2.48	2.36	0.96	0.91	2.06	1.90
Brain	0.02	0.01	0.06	0.04	0.02	0.02	0.06	0.04
Salivary glands	0.08	0.05	0.27	0.17	0.11	0.08	0.32	0.21
Skin	0.36	0.28	0.67	0.55	0.35	0.28	0.66	0.54
Remainder	0.50	0.41	1.21	1.11	0.56	0.49	1.24	1.10
Effective dose (mSv)	0.76	0.64	1.48	1.30	1.06	0.93	1.87	1.71

on the larynx than those centred on the pituitary mainly due to the higher doses received by thyroid, oesophagus and lung. Eye lens doses, also shown in table 1, are higher for scans centred on pituitary because the eyes are within the imaged field. However the partial rotation from gantry angle -40° to $+160^{\circ}$ results in higher doses at the posterior of the head, thus minimizing dose to the eye lens. Figure 4 shows example dose distributions for head and neck CBCT scans in the male ICRP phantom, illustrating the different scan FOVs simulated and the asymmetry of dose due to partial rotation scanning.

Table 2 shows the organ and effective doses simulated in the ICRP phantom for pelvis CBCT scans. Effective doses increase with exposed field length and with scan total mAs. The exposed organs within the FOV also have a significant influence on effective dose. For example 'Pelvis M15' dose is 1.8 times higher for the female than the male phantom. This is principally due to the much higher dose to gonads and colon. A greater proportion of these organs is within the exposed field for the female phantom than for the male phantom, due to their different anatomical positions and the different scan centre locations (uterus and prostate). Figures 5 and 6 show example dose distributions for pelvis CBCT scans in the male and female ICRP phantoms, illustrating the different scan FOVs simulated. The asymmetry of doses between right and left is caused by the gantry starting and stopping with the kV tube on the patient right side.

Tables 3, 4 and 5 show simulated organ and effective doses for CBCT scans of the thorax. Female organ doses are generally slightly higher due to smaller dimensions of the phantom.

Table 5. Organ and effective doses simulated in the ICRP phantoms for breast CBCT scans and LFOV chest CBCT scans. ^aHalf rotation: gantry angle -180° to $+20^{\circ}$. ^cHalf rotation: gantry angle -60° to $+140^{\circ}$.

Preset	Left breast S10	Left breast S20	Right breast S10	Right breast S20	Chest L10	Chest L10	Chest L20	Chest L20
Isocentre	L breast	L breast	R breast	R breast	Lung centre	Lung centre	Lung centre	Lung centre
Sex M/F	F	F	F	F	F	M	F	M
kV	120	120	120	120	120	120	120	120
Collimator	S10	S20	S10	S20	L10	L10	L20	L20
Filter	F1	F1	F1	F1	F1	F1	F1	F1
Total mAs	117.1	117.1	117.1	117.1	264.0	264.0	264.0	264.0
Rotation	half ^c	half ^c	half ^c	half ^c	full	full	full	full
<i>Organ doses (mGy)</i>								
Bone marrow (red)	0.32	0.71	0.30	0.69	0.62	0.52	1.26	1.09
Colon	0.02	0.06	0.03	0.08	0.04	0.11	0.10	0.38
Lung	2.04	2.85	2.04	2.66	3.75	3.02	4.82	3.91
Stomach	0.48	2.44	0.36	1.49	0.56	0.64	2.38	2.34
Breast	2.15	2.36	4.28	4.62	3.78	3.14	4.08	3.43
Gonads	0.01	0.02	0.01	0.01	0.01	0.01	0.03	0.02
Bladder	0.01	0.01	0.01	0.02	0.01	0.01	0.03	0.03
Oesophagus	1.20	2.26	1.06	2.22	2.60	2.00	4.68	3.40
Liver	0.37	1.39	0.73	2.85	0.81	0.80	3.14	2.70
Thyroid	0.32	2.41	0.37	4.59	0.67	0.53	6.27	4.87
Bone surface	1.07	2.16	0.70	1.58	1.97	1.87	3.68	3.46
Brain	0.02	0.05	0.02	0.05	0.04	0.03	0.09	0.07
Salivary glands	0.07	0.22	0.09	0.29	0.15	0.11	0.40	0.27
Skin	0.32	0.59	0.29	0.55	0.54	0.44	0.97	0.80
Remainder	0.46	1.12	0.49	1.12	0.84	0.69	1.76	1.56
Left breast	3.18	3.46	3.41	3.74				
Right breast	1.12	1.26	5.14	5.51				
Effective dose (mSv)	0.75	1.42	1.00	1.69	1.34	1.13	2.35	2.01

Doses increase with field length due to greater scatter and greater number of body organs within the exposed region. The 'Symmetry 4D' preset has the highest dose due to the higher mAs and use of F0 rather than F1 filter. SFOV half rotation effective doses vary by almost a factor of three depending on the laterality of the isocentre, patient sex and imaged field length. Dose to breast from half rotation breast CBCT scans is also highly variable depending on the partial rotation used, ranging from 1.1 mGy to 5.5 mGy. The degree of dose sparing that can be achieved to the contralateral breast is limited by the partial rotation that can be used, which cannot extend past $\pm 180^{\circ}$. This results in the better contralateral breast sparing achieved for left than right breast scans. Figures 7 and 8 show simulated dose distributions from CBCT scans of the chest with MFOV and SFOV respectively.

4. Discussion

The simulated organ and effective doses given here correspond to the default protocols provided in the current version of the Elekta XVI CBCT system (XVI version 5.0). Since organ

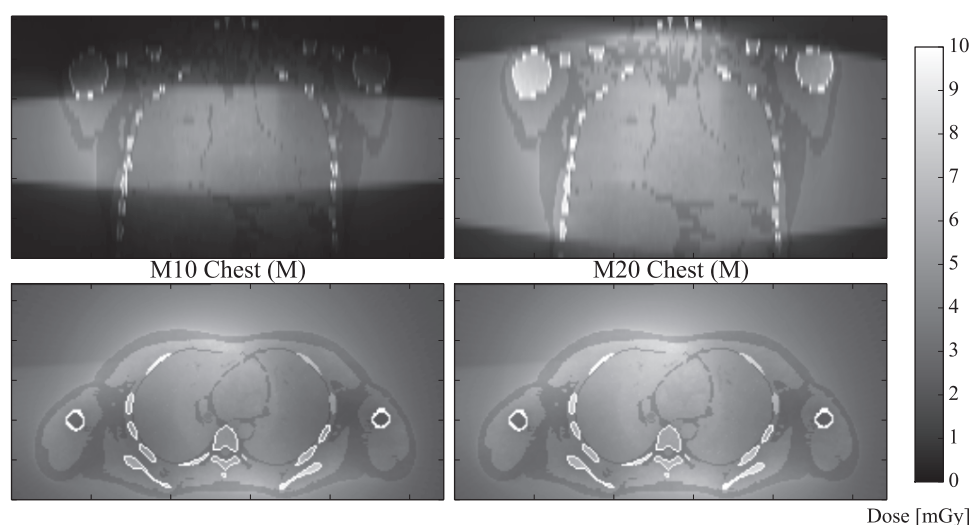


Figure 7. Simulated dose distributions in the ICRP male phantom from MFOV CBCT scans of the chest.

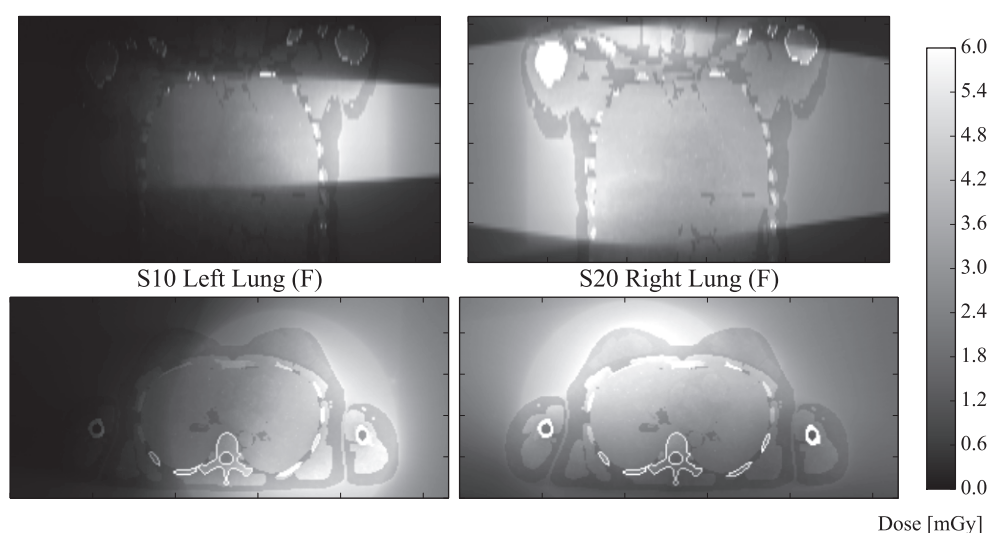


Figure 8. Simulated dose distributions in the ICRP female phantom from SFOV half-rotation CBCT scans of the chest.

and effective doses scale linearly with mAs, the information given here can be used to calculate doses for other protocols using the same combinations of kV, filter, collimator, rotation angle.

Effective doses for Elekta XVI CBCT that have been previously reported in the literature [6, 7, 12, 13] are for protocols which do not correspond exactly to those simulated here. However comparisons can be made where the protocols differ only in total mAs. For a head and neck scan with the S20 collimator and 36 mAs, Hyer *et al* [6] and Dufek *et al* [7] quote effective doses of 0.04 mSv and 0.1 mSv respectively. When scaled appropriately our data give doses of 0.06 mSv and 0.12 mSv for a scans of the male phantom centred on the pituitary

and larynx respectively. Scans of the female phantom resulted in doses about 50% higher. For M10 pelvis scans centred on the prostate Hyer *et al* [6], Dufek *et al* [7], and Halg *et al* [12] give effective doses of 2.3 mSv, 4.1 mSv and 3.1 mSv respectively (values normalized to scan with 1040 mAs). The corresponding dose for an M10 pelvis scan from our simulations is 1.9 mSv. For an M20 chest scan centred on the lungs with 1028 mAs, Hyer *et al* [6] reports effective dose of 7.15 mSv, which compares to our simulated dose of 10.7 mSv for the male ICRP phantom (increasing to 12.4 mSv for the female phantom). Differences between the doses reported here and those published by other authors may be due to a number of causes, including use of different phantoms, errors due to extrapolating mean organ dose from point dose measurements, and differences in dose output of individual XVI systems.

Many studies of CBCT dose present results for a small number of protocols, typically categorized as head, chest and pelvis [5, 6, 9, 11, 13, 22]. However a wide variety of protocol variations may be used in clinical practice in both the anatomical location of the scan, and in acquisition parameters such as collimator, filter and partial rotation gantry angles. Table 2 reveals differences in head scan effective dose of approximately a factor of two between those centred on pituitary and larynx. Effective dose for pelvis scans varies markedly between male and female patients and with the FOV length mostly due to the effect on dose to gonads and colon. Patient dose from chest CBCT scans also varies strongly with FOV length, increasing by a factor of two between M10 and M20 scans. Scan position is also important for chest scans, which may be centred on tumour sites in the left or right lung. A few authors [8, 21, 23, 24] have considered the effect on scan dose for off-centre chest scans and for partial rotation chest protocols, although full organ and effective doses have not been presented. Our simulations show how the gantry start and stop angles for partial rotation scans influence the scan dose. For example the rotation used at our institution for imaging of right lung sites (-180° to $+20^\circ$) results in 25–50% higher effective dose than the rotation used for imaging left lung sites (-80° to $+120^\circ$).

We have simulated CBCT scan doses in the male and female ICRP computational phantoms, which are consistent with reference adult anatomical data from ICRP report 89 [37] and are recommended for studies of radiation dose and determination of effective dose. However it should be remembered that the doses presented here are representative for an average patient, and there will be significant variations for individual patients. CT imaging dose is highly dependent on patient size [38] and a number of authors have noted increased CBCT dose for pediatric patients [17, 20, 39].

CBCT scan doses in this paper are presented as organ and effective doses. It should be acknowledged that use of effective dose to evaluate risk for patients undergoing radiotherapy is non ideal, due to the difficulty of combining with risks from the radiotherapy dose received. However, effective dose is still commonly used to assess IGRT dose burden. The report of AAPM TG-75 [40] recommends effective dose as the only quantity allowing inter-comparison of stochastic risk between different imaging scenarios. Halg *et al* [12] suggested excluding organs receiving radiotherapy dose greater than 5 Gy from the effective dose calculation to better represent the *additional* risk from imaging. Such a calculation would need to be done on an individual patient basis due to patient specific variations in radiotherapy dose distribution. However, the tables of organ doses presented here would allow re-calculation of effective dose based on a subset of organs as required.

In summary the XVI head and neck presets can be considered low dose with effective doses less than 0.1 mSv. XVI 3D Chest presets have intermediate doses ranging from 1.38 to 3.19 mSv. XVI prostate and pelvis presets have generally higher doses ranging from 2.80 to 7.60 mSv. Four-dimensional (Symmetry) lung scans give the highest imaging dose of 7.88 to 8.30 mSv. The simulated CBCT doses reported here can aid selection of appropriate scanning

protocols. For example the clear increase in dose with length of the imaged FOV supports a recommendation to use shorter FOVs whenever possible (e.g. M10 rather than M20 collimator). The expected increase in dose with increased mAs is also observed, for example compare Prostate M15 with Pelvis M15 protocol in table 2. Dose can be minimized by using the M15 Pelvis protocol instead of M15 Prostate. However the trade-off between dose and image quality must be considered [41]; imaging dose should not be reduced below that necessary to produce a clinically acceptable image for the specified purpose.

5. Conclusions

A Monte Carlo model of the Elekta XVI CBCT system has been developed and validated using the Geant4 application for tomographic emission (GATE). The model has been used to simulate patient doses in the ICRP male and female computational phantoms from a comprehensive set of CBCT imaging protocols, including all default protocols in the current version of XVI. Effective doses from head and neck CBCT scans range from 0.03 to 0.09 mSv. Pelvis CBCT scan effective doses range from 2.8 to 7.6 mSv, and medium FOV chest scan doses range from 1.4 to 3.2 mSv. The information presented here will be of use in assessing patient dose from CBCT imaging, as required for justification of imaging procedures. The CBCT MC model developed here may be useful to others for studies of CBCT imaging and dosimetry. Please contact the corresponding author for details of how to obtain the model.

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